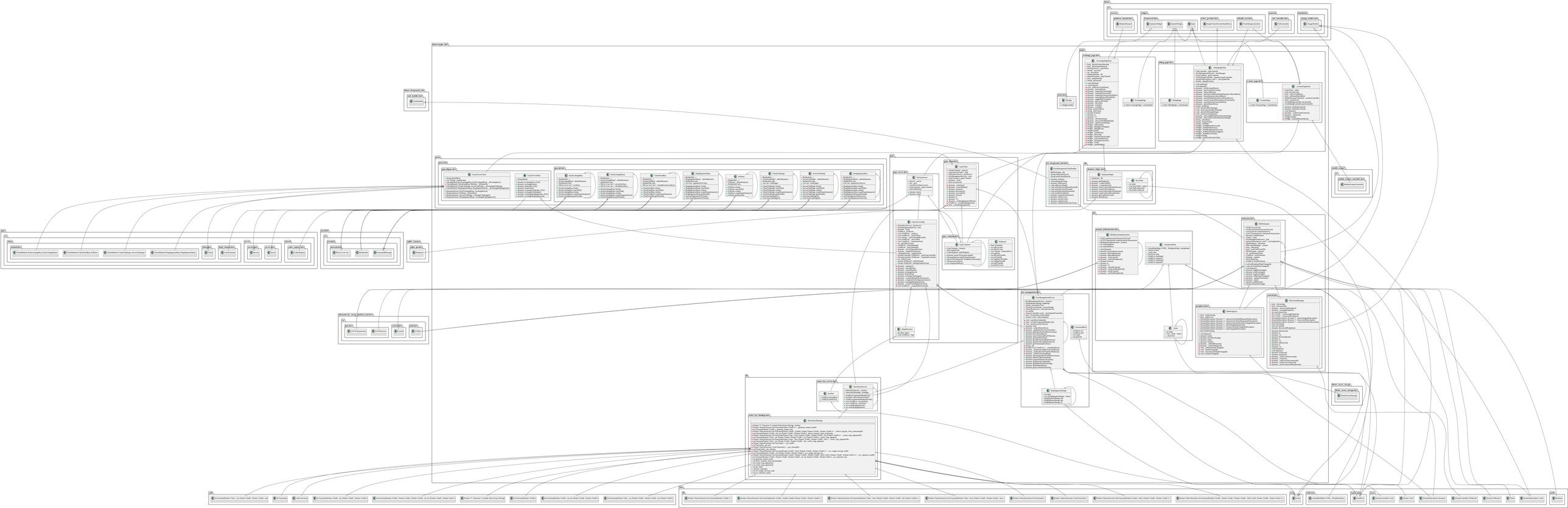

flutter
src
services
platform_channel.dart
MethodChannel
widgets
framework.dart
StatelessWidget
StatefulWidget
State
ticker_provider.dart
SingleTickerProviderStateMixin
editable_text.dart
TextEditingController
material
tab_controller.dart
TabController
foundation
change_notifier.dart
ChangeNotifier
fluttersample_2025
pages
exchange_page.dart
_ExchangePageState
bool _bleAdvertiserRunning
bool _bleScannerRunning
PsiGrpcServer? _grpcServer
String? _serverIp
int _serverPort
DatabaseHelper _db
MethodChannel _mapChannel
bool _grpcRunning
String _qrPayload
void initState()
void dispose()
void _onReceiveTaskData()
dynamic _hasAnyKey()
dynamic _requireKeyWarning()
dynamic _requestPermissions()
dynamic _requestPermissionsSnakBar()
dynamic _ensureBluetoothEnabled()
dynamic _toggleBleExchange()
dynamic _getLocalWifiIp()
dynamic _showQr()
dynamic _stopQr()
dynamic _scanQr()
String _bytesToHex()
String _fmtTime()
dynamic Future()
dynamic ()
dynamic >()
dynamic _calcMeetStats()
dynamic _showUnifiedPsiDialog()
dynamic _openExternalMap()
Widget _mathLabel()
Widget _debugRowWidget()
Widget _debugRow()
Widget build()
Widget _statusRow()
Widget _bleCard()
Widget _familiarCheckCard()
Widget _qrActionButtons()
Widget _qrDisplaySection()
Widget _card()
Widget _statusBadge()
ExchangePage
State<ExchangePage> createState()
debug_page.dart
DebugPage
State<DebugPage> createState()
_DebugPageState
TabController _tabController
KeyManagementService _keyManager
GrpcCommon _grpcCommon
TextEditingController _dummyCountController
StreamSubscription<void>? _keyUpdateSub
double _debugFontSize
void initState()
void dispose()
dynamic _fetchLimitedKeys()
dynamic _fetchTotalKeyCount()
dynamic _deleteAllKeys()
dynamic _generateAndInsertSingleDummyCollectedKey()
dynamic _insertDummyCollectedKey()
dynamic _insertMultipleDummyCollectedKeys()
dynamic _insertCommonDummyKeysFromAsset()
dynamic _insertDummyGeneratedKey()
dynamic _deleteMasterKey()
String _fullHex()
void _showFullKeyDialog()
void _showGeneratedKeyDialog()
void _showFullStringDialog()
void _showErrorSnackbar()
dynamic _showAddMultipleDummiesDialog()
dynamic _showAddCommonDummyDialog()
String _shortHex()
String _formatTime()
String _toDMS()
Widget _buildMasterKeyCard()
Widget _buildSlotSelector()
Widget _buildRingRangeSelector()
Widget _buildGrpcOptionToggles()
Widget _buildKeyListTab()
Widget build()
Widget _buildVerificationTab()
scanner_page.dart
ScannerPage
State<ScannerPage> createState()
_ScannerPageState
GrpcClient _client
bool isConnecting
bool _isProcessingScan
bool _isManualInputMode
MobileScannerController _scannerController
bool _cameraLock
TextEditingController ipController
TextEditingController portController
dynamic safeStopCamera()
dynamic safeStartCamera()
void dispose()
dynamic _confirmAndConnect()
dynamic _onDetect()
Widget build()
Widget _manualBottomSheet()
main.dart
MyApp
Widget build()
key_foreground_task.dart
KeyManagementTaskHandler
BleNickname _ble
String notificationLocal
String notificationRemote
dynamic onStart()
void onRepeatEvent()
dynamic onDestroy()
void onReceiveData()
void onNotificationButtonPressed()
void onNotificationPressed()
void onNotificationDismissed()
void sendDataToMain()
dynamic requestPermissions()
dynamic initService()
dynamic startService()
dynamic stopService()
dynamic UpdateNotificationText()
db
database_helper.dart
DatabaseHelper
Database? _db
dynamic getDatabase()
dynamic _createTables()
dynamic _createIndexes()
dynamic insertCommonDummyKeys()
dynamic existsCollectedKey()
dynamic insertCollectedKeyIfAbsent()
dynamic insertCollectedKeysBatch()
dynamic getTotalKeyCount()
dynamic deleteKey()
KeyTable
int index
List<KeyTable> values
KeyTable generated
KeyTable collected
proto
generated
grpc.pbgrpc.dart
GrpcServiceClient
String defaultHost
List<String> oauthScopes
ClientMethod<KeyExchangeReq, KeyExchangeResp> _$exchangeKeys
ClientMethod<ClientFinalReq, PsiDone> _$finalizePsi
ClientMethod<ClientChallenge, ServerChallenge> _$exchangeChallenges
ClientMethod<RingSignatureReq, RingSignatureResp> _$exchangeRingSignatures
ResponseFuture<KeyExchangeResp> exchangeKeys()
ResponseFuture<PsiDone> finalizePsi()
ResponseFuture<ServerChallenge> exchangeChallenges()
ResponseFuture<RingSignatureResp> exchangeRingSignatures()
GrpcServiceBase
String $name
dynamic exchangeKeys_Pre()
dynamic exchangeKeys()
dynamic finalizePsi_Pre()
dynamic finalizePsi()
dynamic exchangeChallenges_Pre()
dynamic exchangeChallenges()
dynamic exchangeRingSignatures_Pre()
dynamic exchangeRingSignatures()
grpc.pb.dart
KeyExchangeReq
BuilderInfo _i
KeyExchangeReq? _defaultInstance
BuilderInfo info_
PbList<List<int>> encKeys
KeyExchangeReq clone()
KeyExchangeReq copyWith()
KeyExchangeReq create()
KeyExchangeReq createEmptyInstance()
KeyExchangeReq getDefault()
KeyExchangeResp
BuilderInfo _i
KeyExchangeResp? _defaultInstance
BuilderInfo info_
PbList<List<int>> serverEncKeys
PbList<List<int>> clientReencKeys
KeyExchangeResp clone()
KeyExchangeResp copyWith()
KeyExchangeResp create()
KeyExchangeResp createEmptyInstance()
KeyExchangeResp getDefault()
ClientFinalReq
BuilderInfo _i
ClientFinalReq? _defaultInstance
BuilderInfo info_
PbList<List<int>> clientReencServerKeys
ClientFinalReq clone()
ClientFinalReq copyWith()
ClientFinalReq create()
ClientFinalReq createEmptyInstance()
ClientFinalReq getDefault()
RingSignatureResp
BuilderInfo _i
RingSignatureResp? _defaultInstance
BuilderInfo info_
List<int> signatureForClient
RingSignatureResp clone()
RingSignatureResp copyWith()
RingSignatureResp create()
RingSignatureResp createEmptyInstance()
RingSignatureResp getDefault()
bool hasSignatureForClient()
void clearSignatureForClient()
PsiDone
BuilderInfo _i
PsiDone? _defaultInstance
BuilderInfo info_
PsiDone clone()
PsiDone copyWith()
PsiDone create()
PsiDone createEmptyInstance()
PsiDone getDefault()
ClientChallenge
BuilderInfo _i
ClientChallenge? _defaultInstance
BuilderInfo info_
List<int> challengeC
ClientChallenge clone()
ClientChallenge copyWith()
ClientChallenge create()
ClientChallenge createEmptyInstance()
ClientChallenge getDefault()
bool hasChallengeC()
void clearChallengeC()
ServerChallenge
BuilderInfo _i
ServerChallenge? _defaultInstance
BuilderInfo info_
List<int> challengeS
ServerChallenge clone()
ServerChallenge copyWith()
ServerChallenge create()
ServerChallenge createEmptyInstance()
ServerChallenge getDefault()
bool hasChallengeS()
void clearChallengeS()
RingSignatureReq
BuilderInfo _i
RingSignatureReq? _defaultInstance
BuilderInfo info_
List<int> signatureForServer
RingSignatureReq clone()
RingSignatureReq copyWith()
RingSignatureReq create()
RingSignatureReq createEmptyInstance()
RingSignatureReq getDefault()
bool hasSignatureForServer()
void clearSignatureForServer()
grpc
grpc_server.dart
PsiGrpcServer
Server? _server
int? _port
GrpcServiceImpl service
GrpcCommon _grpcCommon
bool isRunning
dynamic start()
dynamic stop()
GrpcServiceImpl
NativeKeyService _keyService
KeyManagementService _kms
dynamic _ready
Uint8List _mySecret
List<Uint8List> _myKeys
List<Uint8List> _myEncKeys
List<String> _myGeneratedKeysHex
List<Uint8List> _serverAbQ
List<Uint8List> _lastIntersection
int _lastSaKeyCount
Uint8List? _lastChallengeC
Uint8List? _lastChallengeS
dynamic _serverSignatureFuture
dynamic _ringSelectionFuture
_RingSelection? _ringSelection
StreamController<PsiResult> _psiEventController
StreamController<PsiResult> _ringAuthController
int _sbKeyCount
Stream<PsiResult> onPsiFinished
Stream<PsiResult> onRingAuthenticated
dynamic _initialize()
dynamic _reloadKeys()
dynamic _ensureReady()
dynamic exchangeKeys()
dynamic finalizePsi()
dynamic exchangeChallenges()
dynamic _computeRingSelectionAsync()
dynamic _computeServerSignatureAsync()
dynamic exchangeRingSignatures()
dynamic _verifyClientSignatureLater()
List<Uint8List> _computeIntersection()
_RingSelection
KeyPair signer
List<Uint8List> ring
grpc_client.dart
GrpcClient
ClientChannel? _channel
GrpcServiceClient? _stub
NativeKeyService _keyService
KeyManagementService _kms
GrpcCommon _grpcCommon
dynamic _ready
bool isConnected
dynamic _initialize()
dynamic _ensureReady()
dynamic connect()
dynamic disconnect()
dynamic executePsi()
dynamic Future()
dynamic ()
dynamic >()
dynamic _runRingSignaturePhase()
Uint8List? _createRingSignature()
bool _verifyRingSignature()
grpc_common.dart
GrpcCommon
GrpcCommon _instance
bool enableGzip
CodecRegistry codecRegistry
dynamic ensureTlsAssetsLoaded()
ChannelOptions buildClientOptions()
ServerTlsCredentials? buildServerSecurity()
String bytesToHex()
int comparePubKey()
PsiResult
bool isFamiliar
int saKeyCount
int sbKeyCount
List<String> commonKeys
int ringSize
int dbLoadTimeMs
int psiTimeMs
int ringSelectTimeMs
int ringSigTimeMs
int totalTimeMs
int psiKeyCount
key_management.dart
KeyManagementService
KeyManagementService _instance
RingSignatureRange ringRange
String _masterKeyAlias
FlutterSecureStorage _secureStorage
NativeKeyService _nativeKeyService
int slotMs
StreamController<void> _keyUpdatedController
bool _locationServicePermitted
Stream<void> onKeyUpdated
void _notifyKeyUpdated()
bool _isValidCompressedPubkey33()
void _attachLocationAsync()
dynamic init()
dynamic _ensureMasterKey()
dynamic _getPublicKeyForAdvertise()
dynamic getPublicKeyForBleAdvertise()
dynamic hasCollectedKey()
dynamic insertCollectedKeyIfAbsent()
dynamic getLatestKeyPair()
dynamic getAllGeneratedPublicKeys()
dynamic getAllCollectedPublicKeys()
dynamic getTimestampForKey()
String _k()
Iterable<List<Uint8List>> _chunkPubkeys()
dynamic _loadGeneratedRowsForPubkeys()
dynamic _loadCollectedTimesForPubkeys()
dynamic _buildTimestampMap()
dynamic selectSignerKeyFromIntersection()
dynamic filterKeysBySameSlot()
KeyPair? generateDummyKeyPair()
dynamic getMasterKeyBase64()
dynamic getMasterKeyHexString()
dynamic deleteMasterKey()
dynamic geoLocatorPermission()
_GeneratedRow
Uint8List sec
Uint8List pub
int expire
int generate
RingSignatureRange
int index
List<RingSignatureRange> values
RingSignatureRange slot
RingSignatureRange day
RingSignatureRange all
ble
mutual_authentication.dart
BleMutualAuthentication
UUID authenticationCharacteristicUuid
GATTCharacteristic authenticationCharacteristic
BleMutualAuthentication _instance
int challengeBytes
int responseBytes
void dispose()
dynamic startAuthentication()
dynamic onWriteRequest()
dynamic onReadRequest()
dynamic _firstCentral()
dynamic _onFirstPeripheral()
dynamic Future()
dynamic ()
dynamic >()
dynamic _secondCentral()
dynamic _onSecondPeripheral()
dynamic _thirdCentral()
dynamic _onThirdPeripheral()
_PeripheralState
LinkedHashMap<UUID, _PeripheralState> peripherals
Central central
_State state
DateTime time
Uint8List challenge1
Uint8List response1
Uint8List challenge2
Uint8List response2
_State
int index
List<_State> values
_State first
nickname.dart
BleNickname
UUID serviceUuid
UUID nicknameCharacteristicUuid
GATTService nicknameService
GATTCharacteristic nicknameCharacteristic
Duration validDuration
Timer? _timer
KeyManagementService _kms
StreamSubscription<void>? _keyUpdateSub
BlePeripheral _advertiser
BleCentralManager _scanner
bool _isRunning
bool _waitForPoweredOn
BleNickname _instance
int _lastSlotStartTime
Uint8List _lastNickname
Random _random
bool isRunning
Uint8List localNickname
void onPeripheralStateChanged()
void onCentralStateChanged()
void dispose()
dynamic toggleExchange()
dynamic startExchange()
dynamic stopExchange()
dynamic onBleStateChanged()
dynamic _updateNickname()
dynamic _start()
dynamic addRemote()
String nickname2string()
peripheral.dart
BlePeripheral
bool _isAdvertising
bool _addService
StreamSubscription<dynamic>? _characteristicReadRequestedSubscription
StreamSubscription<dynamic>? _characteristicWriteRequestedSubscription
StreamSubscription<dynamic>? _characteristicNotifyStateChangedSubscription
StreamSubscription<dynamic>? _mtuChangedSubscription
StreamSubscription<dynamic>? _connectionStateChangedSubscription
StreamSubscription<dynamic>? _stateChangedSubscription
bool isAdvertising
void dispose()
dynamic start()
dynamic startAdvertising()
dynamic stop()
dynamic restart()
dynamic _onReadRequest()
dynamic _onWriteRequest()
void _onNotifyStateChanged()
void _onMtuChanged()
void _onConnectionStateChanged()
void _onStateChanged()
central.dart
BleCentralManager
bool _isScanning
bool _isPoweredOn
dynamic _discoveredPeripherals
dynamic _waitingPeripherals
int maxConnections
Set<UUID> _connectingPeripherals
Set<UUID> _connectedPeripherals
StreamSubscription<dynamic>? _stateChangedSubscription
StreamSubscription<dynamic>? _discoveredSubscription
StreamSubscription<dynamic>? _connectionStateSubscription
bool isScanning
bool isAvailable
dynamic discoveredPeripherals
dynamic QueueList()
dynamic ()
dynamic >()
dynamic PriorityQueue()
dynamic ()
dynamic >()
dynamic QueueList()
dynamic ()
dynamic >()
void initialize()
void dispose()
dynamic startScan()
dynamic stopScan()
dynamic _onDeviceDiscovered()
dynamic _connect()
dynamic _onDeviceDisconnected()
dynamic _onDeviceConnected()
dynamic _printConnectedPeripherals()
ffi
native_key_service.dart
KeyPair
Uint8List privateKey
Uint8List publicKey
NativeKeyService
NativeKeyService _instance
NativeKeyBindings _bindings
Uint8List? generateMasterKey()
KeyPair? deriveNewKeyPair()
Uint8List generateRandomSecret()
List<Uint8List> encryptSet()
List<Uint8List> intersect()
int createRingSignature()
int verifyRingSignature()
native_key_bindings.dart
NativeKeyBindings
Pointer<T> Function<T extends NativeType>(String) _lookup
Pointer<NativeFunction<Int Function(Pointer<Uint8>)>> _generate_master_keyPtr
int Function(Pointer<Uint8>) _generate_master_key
Pointer<NativeFunction<Int Function(Pointer<Uint8>, Uint64, Uint64, Pointer<Uint8>, Pointer<Uint8>)>> _derive_keypair_from_timestampPtr
int Function(Pointer<Uint8>, int, int, Pointer<Uint8>, Pointer<Uint8>) _derive_keypair_from_timestamp
Pointer<NativeFunction<Int Function(Pointer<Char>, Size, Pointer<Uint8>, Pointer<Uint8>, Int, Pointer<Uint8>)>> _create_ring_signaturePtr
int Function(Pointer<Char>, int, Pointer<Uint8>, Pointer<Uint8>, int, Pointer<Uint8>) _create_ring_signature
Pointer<NativeFunction<Int Function(Pointer<Char>, Size, Pointer<Uint8>, Pointer<Uint8>, Int)>> _verify_ring_signaturePtr
int Function(Pointer<Char>, int, Pointer<Uint8>, Pointer<Uint8>, int) _verify_ring_signature
Pointer<NativeFunction<Int Function()>> _psi_initPtr
int Function() _psi_init
Pointer<NativeFunction<Void Function()>> _psi_cleanupPtr
void Function() _psi_cleanup
Pointer<NativeFunction<Int Function(Pointer<Uint8>, Int32, Pointer<Uint8>, Pointer<Uint8>)>> _ecc_single_encrypt_setPtr
int Function(Pointer<Uint8>, int, Pointer<Uint8>, Pointer<Uint8>) _ecc_single_encrypt_set
Pointer<NativeFunction<Int Function(Pointer<Uint8>, Pointer<Uint8>, Pointer<Uint8>, Int32, Int32, Pointer<Uint8>, Pointer<Int32>)>> _ecc_intersect_setsPtr
int Function(Pointer<Uint8>, Pointer<Uint8>, Pointer<Uint8>, int, int, Pointer<Uint8>, Pointer<Int32>) _ecc_intersect_sets
int generate_master_key()
int derive_keypair_from_timestamp()
int create_ring_signature()
int verify_ring_signature()
int psi_init()
void psi_cleanup()
int ecc_single_encrypt_set()
int ecc_intersect_sets()
flutter_foreground_task
task_handler.dart
TaskHandler
grpc
src
client
method.dart
ClientMethod<KeyExchangeReq, KeyExchangeResp>
ClientMethod<ClientFinalReq, PsiDone>
ClientMethod<ClientChallenge, ServerChallenge>
ClientMethod<RingSignatureReq, RingSignatureResp>
client.dart
Client
http2_channel.dart
ClientChannel
server
service.dart
Service
server.dart
Server
shared
codec_registry.dart
CodecRegistry
protobuf
src
protobuf
internal.dart
PbList<List<int>>
BuilderInfo
GeneratedMessage
sqflite_common
sqlite_api.dart
Database
bluetooth_low_energy_platform_interface
src
gatt.dart
GATTCharacteristic
GATTService
central.dart
Central
uuid.dart
UUID
mobile_scanner
src
mobile_scanner_controller.dart
MobileScannerController
flutter_secure_storage
flutter_secure_storage.dart
FlutterSecureStorage
null
int Function(Pointer<Char>, int, Pointer<Uint8>, Pointer<Uint8>, int)
int Function()
void Function()
int Function(Pointer<Uint8>, int, Pointer<Uint8>, Pointer<Uint8>)
int Function(Pointer<Uint8>, Pointer<Uint8>, Pointer<Uint8>, int, int, Pointer<Uint8>, Pointer<Int32>)
Pointer<T> Function<T extends NativeType>(String)
int Function(Pointer<Uint8>)
int Function(Pointer<Uint8>, int, int, Pointer<Uint8>, Pointer<Uint8>)
int Function(Pointer<Char>, int, Pointer<Uint8>, Pointer<Uint8>, int, Pointer<Uint8>)
dart
ffi
Pointer<NativeFunction<Int Function(Pointer<Uint8>)>>
Pointer<NativeFunction<Int Function(Pointer<Uint8>, Uint64, Uint64, Pointer<Uint8>, Pointer<Uint8>)>>
Pointer<NativeFunction<Int Function(Pointer<Char>, Size, Pointer<Uint8>, Pointer<Uint8>, Int, Pointer<Uint8>)>>
Pointer<NativeFunction<Int Function(Pointer<Char>, Size, Pointer<Uint8>, Pointer<Uint8>, Int)>>
Pointer<NativeFunction<Int Function()>>
Pointer<NativeFunction<Void Function()>>
Pointer<NativeFunction<Int Function(Pointer<Uint8>, Int32, Pointer<Uint8>, Pointer<Uint8>)>>
Pointer<NativeFunction<Int Function(Pointer<Uint8>, Pointer<Uint8>, Pointer<Uint8>, Int32, Int32, Pointer<Uint8>, Pointer<Int32>)>>
core
Enum
collection
LinkedHashMap<UUID, _PeripheralState>
typed_data
Uint8List
async
StreamController<void>
Stream<void>
StreamSubscription<dynamic>
StreamController<PsiResult>
Stream<PsiResult>
Timer
StreamSubscription<void>
math
Random